

SUMMARY

Data Science and Engineering undergraduate specializing in full-stack development, scalable backend architecture, and secure system design for high-traffic platforms. Strong analytical problem solver recognized through competitive programming achievements (Codeforces Specialist, 750+ LeetCode solutions), highly dedicated to building efficient, user-centric software solutions that drive measurable business impact.

EDUCATION

National Institute of Technology (NIT), Jalandhar	2023 – Present
<i>B.Tech. in Data Science and Engineering Coursework: ML, NLP, Systems Design, Cloud</i>	CGPA: 8.29
Gobindgarh Public School, Mohali	2021 – 2023
<i>Class 11–12 (Science Stream)</i>	89%

SKILLS

- Programming Languages:** Python, Golang, C, C++, JavaScript, TypeScript
- Web & Full-Stack:** React.js, Next.js, Node.js, Express.js, Tailwind CSS, WebSockets, tRPC, gRPC, Prisma ORM
- Cloud, Databases & Tools:** PostgreSQL, MongoDB, MySQL, Docker, Kubernetes, AWS, Linux, Git, REST APIs, Microservices
- AI, ML & Data Science:** Deep Learning (CNN/RNN), NLP, TensorFlow, Scikit-learn, OpenCV, Pandas, LangChain, LangGraph, Agentic AI, RAG Systems, Vector Databases
- Core Competencies:** Data Structures, Object-Oriented Programming (OOP), DBMS, System Design, Operating Systems

EXPERIENCE

- Advance Application Engineering (AEH) Intern — AI Agent Development — Accenture, Bangalore — Jun 2026–Present:**
 - Architected enterprise **AI agents** via **RAG** pipelines to ground LLMs in proprietary data.
 - Managed **vector embeddings** for semantic search, boosting retrieval accuracy and reducing hallucinations.
 - Built **Node.js/Python** workflows and **React.js/Tailwind CSS** dashboards for real-time monitoring.
- CTP (NITJ Placement Portal) [Live Site] — Backend Lead — NIT Jalandhar — 2024–Present:**
 - Built **Node.js/Express.js** backend, serving **1,200+ users** via **25+ APIs** for automated scheduling.
 - Optimized **MongoDB** and implemented RBAC, boosting retrieval by **35%** and handling **10k+ daily requests**.
 - Engineered **React.js/Tailwind CSS** dashboards, accelerating user navigation by **20%**.
- Xceed (NITJ Timetable) [Live Site] — Top Contributor — NIT Jalandhar — 2025–Present:**
 - Engineered a **React.js/Tailwind CSS** scheduling engine, automating timetables for **5,000+ users** and cutting conflicts by **98%**.
 - Reduced API latency by **40%** via **Express.js** and **MongoDB** optimizations across **10+ departments**.
- Third Space Travel (Travel Booking Platform) [Live Site] — Full-Stack Developer — On-Site — 2026:**
 - Built a **React/TypeScript** travel platform, integrating **30+ APIs** and **Prisma/MySQL** to cut query time by **50ms**.
 - Integrated **PhonePe** SDK and **tRPC** for secure payments, achieving a **99.9% success rate** and reducing drop-offs by **20%**.
 - Developed responsive **SCSS/Framer Motion** components, increasing mobile retention by **25%** and scoring **95/100** on Lighthouse.

PROJECTS

- Shadow-Room: E2EE Secure Chat System [Live Demo] [Repo] — Sole Developer — 2026:**
 - Built an end-to-end encrypted (E2EE) messaging platform utilizing **Golang, React.js**, and **WebSockets** for low-latency, real-time asynchronous communication across distributed networks.
 - Implemented cryptographic handshakes (**RSA, MD5, SHA-256**) and the **Double Ratchet Algorithm** to establish forward secrecy, message integrity, and group scaling via Sender Keys.
 - Integrated multi-factor authentication and executed **Protocol Buffers** modifications to support high-throughput concurrent user sessions while ensuring zero data leaks.
- Secure Token Allocation System [Live Demo] [Repo] — Lead Developer — NIT Jalandhar — 2026:**
 - Developed a cryptographically secure token generation and tracking system to automate physical reporting for **1,200+ incoming students**.
 - Implemented **AES-256** encryption and secure hashing to prevent token forgery and ensure a tamper-proof queue management process.
 - Built a responsive **React.js** dashboard to monitor live allocations, reducing manual verification bottlenecks and wait times by **60%**.
- DataPulse: Operations Management Hub [Live Demo] [Repo] — Full-Stack Developer — 2025:**
 - Engineered a scalable data operations API utilizing **Flask** and **GraphQL** to aggregate and serve complex datasets dynamically.
 - Designed custom **GraphQL** resolvers to eliminate data over-fetching, reducing frontend API payload sizes by **70%**.
 - Integrated **Redis** caching with a **PostgreSQL** database to process high-frequency queries, maintaining a **200ms** response time.

ACHIEVEMENTS

- Speed Quant Competition (First Place):** Secured **1st Rank** among **200+ participants** at NIT Jalandhar for solving complex algorithmic challenges under strict timed conditions.
- Competitive Programming:** Codeforces Specialist (**1502 Rating**, Global Rank 676 in Div. 2) and LeetCode **Top 5%** globally with **1,050+ total algorithmic problems** solved across both platforms.
- AWS Builder Group: Technical Lead** for NIT Jalandhar’s official student community, driving technical workshops on Cloud Computing, DevOps, and AI.